MAHARSHI OZA

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PROFESSIONAL SUMMARY

RTL design engineer with a full front-to-back ASIC background: synthesizable SystemVerilog/Verilog microarchitecture, low-power design (IEEE 1801/UPF), scan-based DFT, and physical design with Design Compiler and ICC2. Experienced in STA-driven timing closure and Python/Makefile flow automation.

CORE COMPETENCIES

Languages & Protocols: SystemVerilog, Verilog HDL, SystemC, C++, Python, Tcl, Makefile, Perl, Shell Scripting, AXI4, I2C, RISC-V ISA
RTL & Low Power: RTL Design, FSM & Datapath, Pipelining, SRAM Integration, Floating-Point Arithmetic, SVA, Clock Domain Crossing (CDC), IEEE 1801/UPF, Power Domains, Isolation, Level Shifters, Clock & Power Gating, High-Level Synthesis
Physical Design & DFT: Synthesis, Floorplanning, Power Grid, Placement, CTS, Routing, Static Timing Analysis (STA), Timing Closure, DRC, Scan Insertion (Muxed Scan), Scan Chains, ATPG Flow
Tools: Synopsys Design Compiler, ICC2, Cadence Genus, Modus, Conformal Low Power, Catapult HLS, VCS, Xcelium, QuestaSim, Vivado, Git, Linux
Computer Architecture: RISC-V, Pipelining, Out-of-Order Execution, Branch Prediction, Memory Hierarchy

PROFESSIONAL EXPERIENCE

RTL Design Verification Engineer | Maxgen Technologies Pvt Ltd., India **Aug 2023 – Aug 2024**

- Contributed to the RTL design of a multi-port packet router in Verilog (FSM control, internal FIFOs, synchronizers, output arbitration); supported synthesis and timing closure in Vivado.
- Implemented an out-of-order AXI4 scoreboard in SystemVerilog/UVM and developed functional coverage reaching over 90%; debugged RTL issues with Questa, Xcelium, VCS, and Vivado.

Functions Engineer – Vision & Perception | Magna Electronics, India **Dec 2021 – Aug 2023**

- Developed validation code for ADAS 2+ lane-marker detection, checking lane width, inter-lane distance, and related geometry against ground truth, and built KPI evaluation utilities for safety-critical perception features.
- Implemented code, branch, and line coverage for 2 perception modules and the KPI framework to expose validation gaps; root-caused edge-case failures with GDB and applied MISRA C practices in development and review.

PROJECTS

Set-Associative Write-Back Cache Controller | SystemVerilog RTL, SVA

- Designed a parameterized N-way set-associative write-back/write-allocate blocking cache controller with selectable LRU/LFU replacement, per-line dirty tracking, a flush/invalidate engine, and valid/ready CPU and memory interfaces with error propagation.
- Partitioned the design into tag-compare, replacement, dirty-tracking, and single-port SRAM modules under a central FSM; wrote a bound SVA checker for handshake and FSM invariants and verified the design in a UVM environment with 0 data mismatches.

Floating-Point Matrix Multiplier | SystemVerilog, Design Compiler

- Designed an SRAM-based single-precision (32-bit) floating-point matrix multiplier around a DesignWare floating-point MAC with FSM-driven address generation and write-back; passed 256/256 result checks across 4 test cases and closed timing at 10 ns in Design Compiler (Nangate 45 nm, ~12.1K cells, ~13.7K μm^2).

High-Performance Transformer Neural Network with Pipelined Self-Attention | SystemVerilog, Design Compiler

- Designed a pipelined Transformer self-attention accelerator computing Query/Key/Value projections, score and attention matrices from SRAM-stored inputs and weights; passed 217/217 result checks across 4 test cases, closed timing at a 5 ns clock in Nangate 45 nm (~13.7K cells, ~14.7K μm^2), achieving an area–performance efficiency of 3.97×10^{-9} $ns^{-1} \cdot \mu m^{-2}$.

Physical Design Space Exploration – ISCAS-89 s38584 | Design Compiler, ICC2, GF180MCU, Python, Makefile, Tcl

- Took the ISCAS-89 s38584 benchmark (~1,250 flops, ~7K cells) through synthesis, floorplan and PG mesh, placement, CTS, routing and route optimization on a GF180MCU 7-track library, with SDC constraints and a single-chain scan configuration.
- Automated the flow with Makefiles and Python, running parallel sweeps of 85 runs (~15–20 min each) over clock period, utilization, max routing layer, clock transition and hold uncertainty; extended the Python report parser to collect 45 metrics per run (WHS, DRCs, area, wirelength, power) and added custom STA, critical-path and ADL pass/fail checks.
- Selected a final configuration (5 ns clock, 0.8 utilization, Metal3 routing) that closed timing with 0 DRCs and positive WHS at a 4.02 ns critical path, ~256.8K μm^2 cell area and ~94% average utilization, reducing the area × delay × routing-layer metric from 17.7M to 3.1M (~5.7×).

Low-Power Design with IEEE 1801 (Cadence course labs) | UPF, Conformal Low Power

- Wrote IEEE 1801 power intent (3 power domains, power switch, isolation cells, 0.9 V → 1.08 V level shifter) and checked it with Conformal Low Power.

Scan Insertion & ATPG Setup: 8-bit Counter (Cadence DFT Fundamentals labs) | Genus, Modus, Verilog, Icarus

- Inserted mux-D scan into an 8-bit counter in Cadence Genus (DFT DRC, one auto-stitched chain) and generated the scan netlist, SCANDEF and Modus ATPG scripts.
- Verified scan shift/capture on the gate-level netlist with a self-checking Verilog testbench (load → capture → unload).

FIR Filter Accelerator via High-Level Synthesis | SystemC, Catapult HLS, RISC-V Rocket-Chip

- Implemented a 16-tap FIR filter accelerator in SystemC and synthesized it with Catapult HLS at a 5 ns clock, exploring architectures that reduced latency from ~3.5K to 69 cycles.

EDUCATION

M.S. Computer Engineering | North Carolina State University, Raleigh, NC | CGPA: 3.7/4 **Aug 2024 – May 2026**
Relevant Coursework: ASIC and FPGA Design with Verilog, Physical Design, Hardware–Software Codesign, Advanced Microprocessor Architecture, ASIC Verification, Advanced Verification using UVM
Training: Cadence – Fundamentals of IEEE 1801 Low-Power Specification Format (v12.0); Design for Test Fundamentals (v25.1)
B.Tech. Electronics & Communication Engineering | Dharmsinh Desai University, India **Aug 2018 – May 2022**